

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459248.8851 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.013
Transit depth (Rp/Rs)^2	3.91 +/- 0.53 [%]
Orbital Inclination (inc)	87.08 +/- 1.4
Airmass coefficient 1 (a1)	0.857 +/- 0.02
Airmass coefficient 2 (a2)	0.11 +/- 0.15
Scatter in the residuals of the lightcurve fit is	2.25 %
Best Comparison Star	#3 - [291, 105]
Optimal Aperture	3.08
Optimal Annulus	12.6
Transit Duration (day)	0.0398 +/- 0.002

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.198 ± 0.013 vs 0.1594 (deviation 2.48σ)
Duration fit vs published	0.0398 d vs 0.0512 d (deviation 4.77σ)
Mid-time O–C	-0.7 min
Depth SNR	12.7
Residual scatter	2.25 %

Light curve

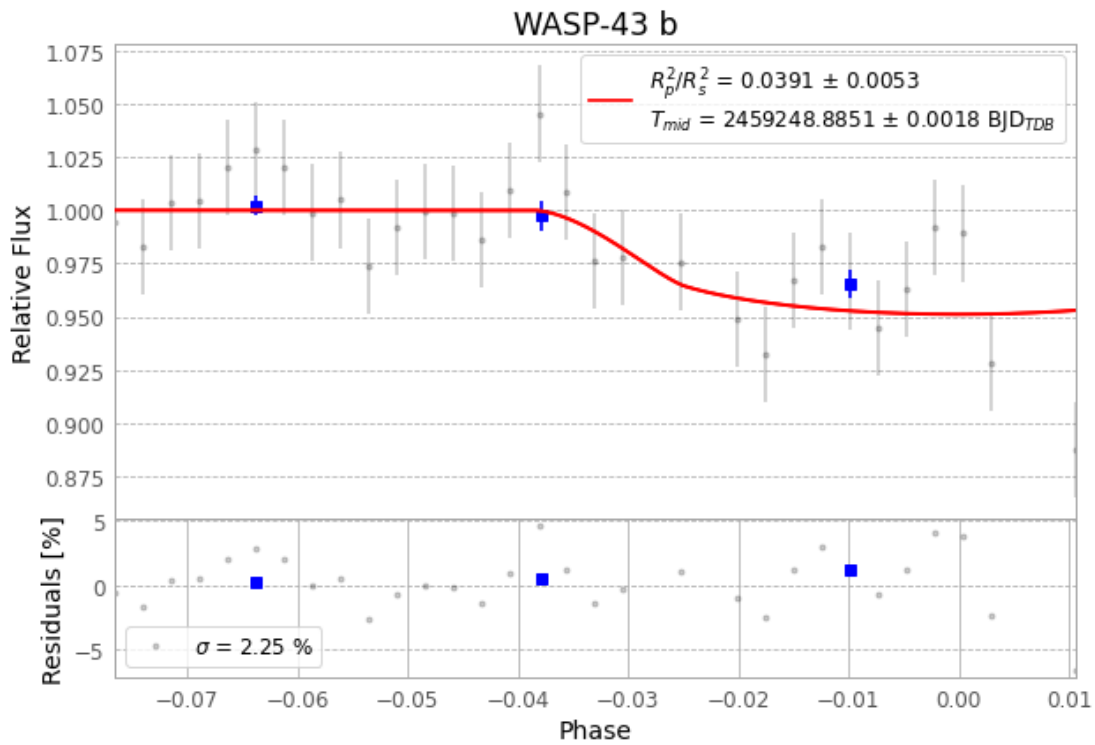


Figure 1 — FinalLightCurve_WASP-43 b_2021-02-03.png (SHA-256 dfe278f6ced78501...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [375, 276], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 42417a90ddf6054d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

63 FITS frames (41 MB), WASP-43210203073612.FITS → WASP-43210203104212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-210203.log (712972d17fb1...), FinalLightCurve_WASP-43 b_2021-02-03.png (dfe278f6ced7...), FinalLightCurve_WASP-43 b_2021-02-03.csv (de1403a10b7c...)

Compute resources

Wall time 315.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 03:18Z.